THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 11: **Keeping Time with the Skies** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A seasonal star appears first in the east before sunrise. Over the next weeks it will appear: 1

(a) not at all **(b)** later in the day only
(c) earlier in the night **(d)** in the west

2. Assertion (A): Lunar calendars need an extra month occasionally to remain aligned with the seasons. Reason (R): A lunar year is about 11 days shorter than a solar year. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. The apparent path of the Sun in the sky at noon is highest in: 1

(a) spring **(b)** winter **(c)** summer **(d)** autumn

4. Which of the following is the best proof that the Earth is rotating? 1

(a) the Moon's phases **(b)** the seasons
(c) the eclipse of the Sun **(d)** daily rise and set of Sun and stars

5. Assertion (A): Festivals following the lunar calendar fall on different Gregorian dates each year. Reason (R): The lunar year and the solar year differ in length. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. The Pole Star's altitude at Jaipur (latitude about 27 degrees N) is about: 1

(a) 90 degrees **(b)** 27 degrees **(c)** 63 degrees **(d)** 0 degrees

7. What is common between a lunar eclipse and an eclipse of the Sun? 1

(a) occurs every full moon **(b)** visible all over the Earth
(c) Sun, Earth and Moon are nearly in a line **(d)** needs special filters to see

8. Assertion (A): The Pole Star is not visible from the southern hemisphere. Reason (R): It lies below the northern horizon there. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. A person standing on the equator sees the Sun at noon on 21 March: 1

(a) overhead **(b)** at the horizon **(c)** at 45 degrees **(d)** not at all

10. At the Arctic Circle in mid-summer, the Sun: 1

(a) never rises **(b)** does not set for a day

(c) rises in the west **(d)** sets at noon

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do all the stars except the Pole Star rise and set? 2

12. Why do we see star trails in a long exposure photograph? 2

13. Find the number of days from 1 January to 15 April in a leap year. 2

14. Why does the length of the day change more rapidly in March and September than in June and December? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe how an observer can show that the stars appear to rotate about the Pole Star in one night using a fixed pole and a window. 3

16. Explain why the lunar eclipse occurs only on a full moon and how long it may last. 3

17. Why were the temples and towers of old India often aligned with the cardinal directions? 3

OR

Explain, using the revolution of the Earth, why the Sun appears against a different constellation each month. 3

18. Explain why Indian astronomers named days of the week after the Sun, the Moon and planets. 3

OR

Explain why the day-night cycle on Mars (24.6 hours) or the Moon (about 29.5 days) differs from the Earth's. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how stars, the Sun and the Moon appear to move across the sky at the equator and at the poles in a day. Why does the Pole Star appear at different heights? 5

OR

Explain how a total solar eclipse occurs, with the geometry, why it occurs rarely at a place, the safety precautions and the observations during totality such as corona and diamond ring. 5

20. Explain the contributions of Indian astronomers (Aryabhata, Varahamihira, Brahmagupta, Sawai Jai Singh II) to time measurement and astronomy. 5

OR

Plan a project to make a calendar of the Moon's phases for one month by observation, including data collection and presentation. 5

END OF QUESTION PAPER — Total: 40 Marks